



Pattern Recognition and Machine Learning

(模式识别与机器学习)

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课程内容

- 绪论
- ~~贝叶斯学习~~
 - 贝叶斯公式
 - ~~贝叶斯判别准则~~
 - **NaiveBayes**（朴素贝叶斯）、隐马尔可夫模型
- 线性分类器
- 非线性分类器
- 特征提取
 - 特征选择
 - PCA/LDA

课程内容

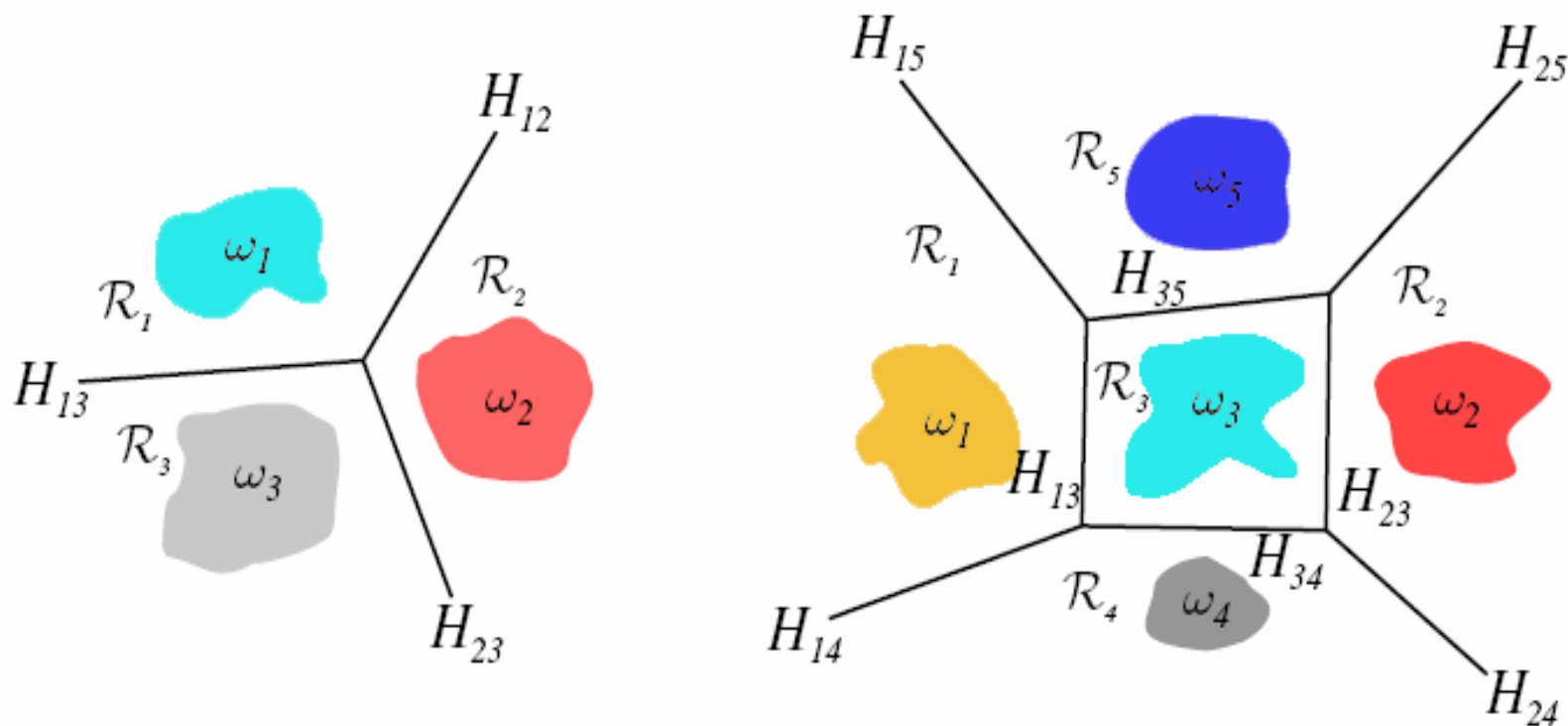
- 浅层->深度学习
 - 随机优化算法
 - 激活函数
 - BN
 - ~~Dropout~~
- 卷积神经网络
- ~~循环神经网络~~
- ~~Transformer~~

主要内容

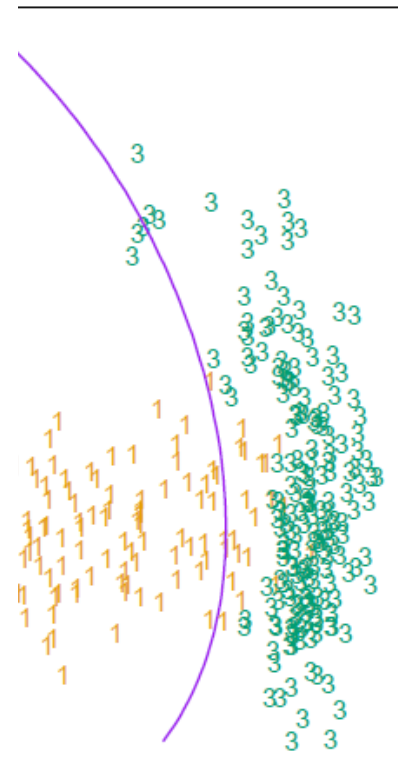
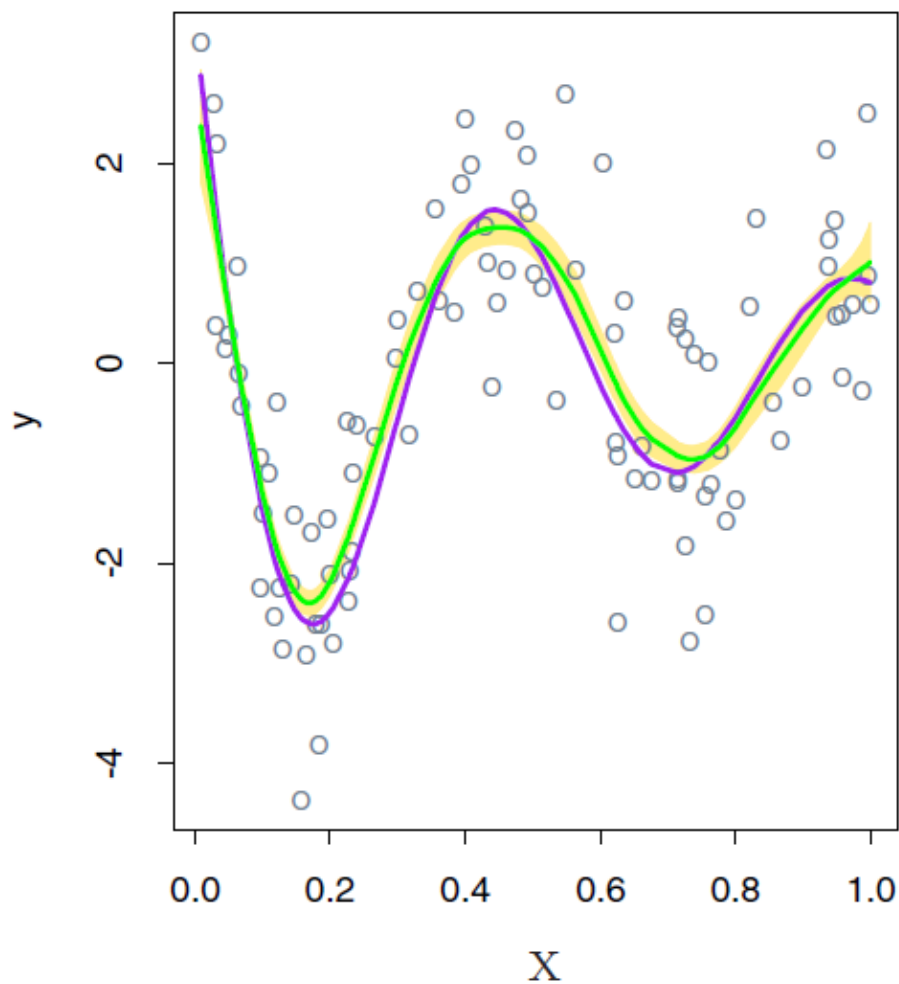
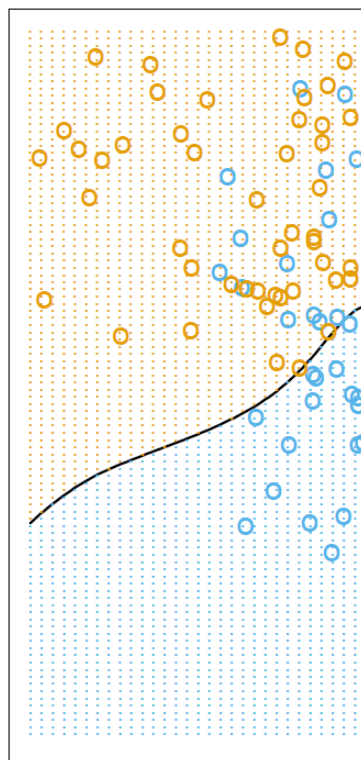
- 机器学习与计算机/人工智能
- 机器学习的定义与内涵
- 机器学习模型与理论
- 学习类型
- 深度学习
- 发展与应用

分类问题

- 高维数据、线性分类面



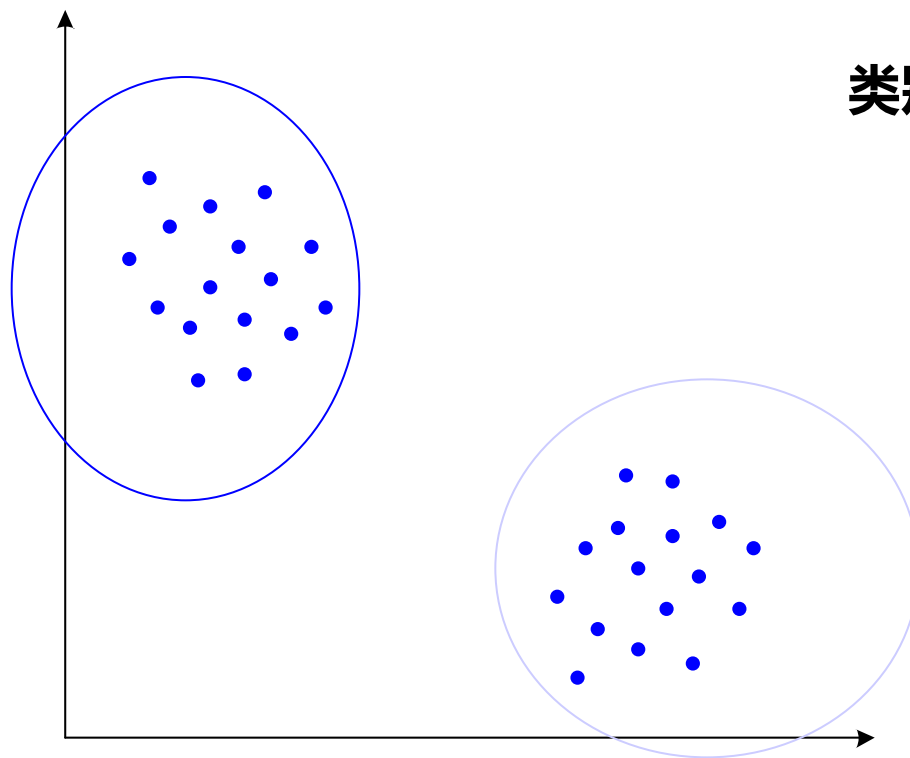
非线性分类器、非线性回归



聚类分析



聚类分析：准则和优化



类别数 $c = 2$

问题的引出

- 两类问题
- 训练数据 $\{(\mathbf{x}_i, y_i) | i = 1, 2, \dots, N_{\text{train}}\}$
- 测试数据 $\{(\mathbf{x}_j, y_j) | j = 1, 2, \dots, N_{\text{test}}\}$
- 分类器 $y = f(\mathbf{x}; \Theta)$
- 如何学习 Θ

$$\Theta = \arg \min \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} \{y_i \neq f(\mathbf{x}_i; \Theta)\}$$

- 问题：
 - 1. 无法求导 $\rightarrow$ proxy loss (代理损失)
 - 2. 过拟合 $\rightarrow$ 正则化 (如: f 应尽可能简单)

如何从训练数据中学习一个分类器

相关性评价

- 独立性

$$P(X, Y) = P(X) P(Y)$$

- 相关性：互信息

$$\begin{aligned} \text{MI}(X, Y) &= \int P(X, Y) \log \frac{P(X, Y)}{P(X)P(Y)} dX dY \\ &= \text{KL}(P(X, Y) \parallel P(X)P(Y)) \end{aligned}$$

Kullback-Leibler散度

- 熵
$$H(X) \equiv \sum_{x \in \mathcal{A}_X} P(x) \log \frac{1}{P(x)},$$

- 相互熵
$$H(X, Y) = \sum_{xy \in \mathcal{A}_X \mathcal{A}_Y} P(x, y) \log \frac{1}{P(x, y)}.$$

- KL散度
$$D_{\text{KL}}(P||Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}.$$

$$I(f_1, f_0) \neq I(f_0, f_1)$$

- Jenson-Shannon熵

$$I_J(f_1, f_0) = I(f_1, f_0) + I(f_0, f_1)$$

Logistic回归目标函数

- 最大似然估计
- 对数似然函数

$$\begin{aligned}\log(L(\mathbf{w})) &= \sum_{i=1}^N \log(P(y_i | \mathbf{x}_i, \mathbf{w})) \\ &= \sum_{i=1}^N \left\{ y_i \mathbf{w}^t \mathbf{x}_i - \log[1 + \exp(\mathbf{w}^t \mathbf{x}_i)] \right\}\end{aligned}$$

- 标注: $\{(y_1=1, y_1=0), \dots\}$
- 预测: $\{(p_1, 1-p_1), \dots\}$

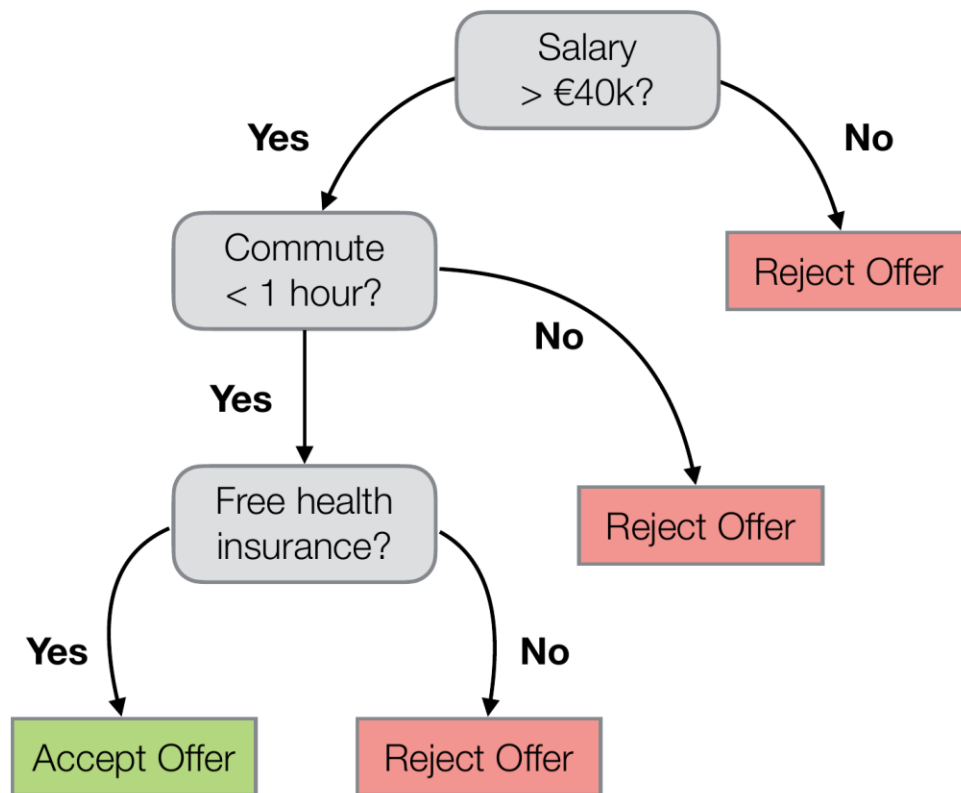
Decision Tree Learning

- Create a classification model to predict class labels by learning decision rules inferred from training data.



Decision:

Should I accept the job offer?



Decision Tree Learning

- **Basic Idea:** Build a tree model that splits the training set into subsets in which all examples have the same class.
- Splitting is done using rules inferred from the training set.
- Each rule is based on a feature, and the split corresponds to the values it can take:
 - e.g. `insured = {true, false}`
 - e.g. `wind = {weak, strong}`
 - e.g. `income = {low, average, high}`
 - e.g. `height < 6ft, height ≥ 6ft`
- If necessary, each subset can be split again using another feature, and so on until all examples have the same class.
- Once the tree is built, we can use it to quickly classify new input examples - this is an eager learning strategy.

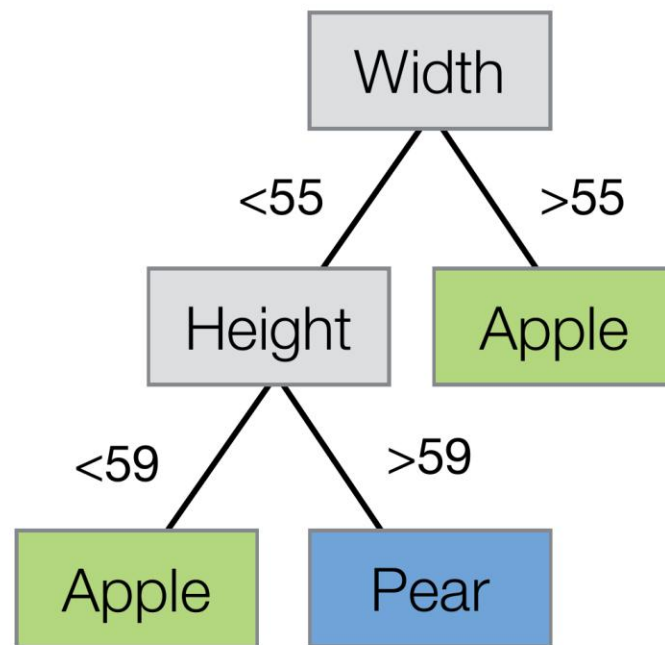
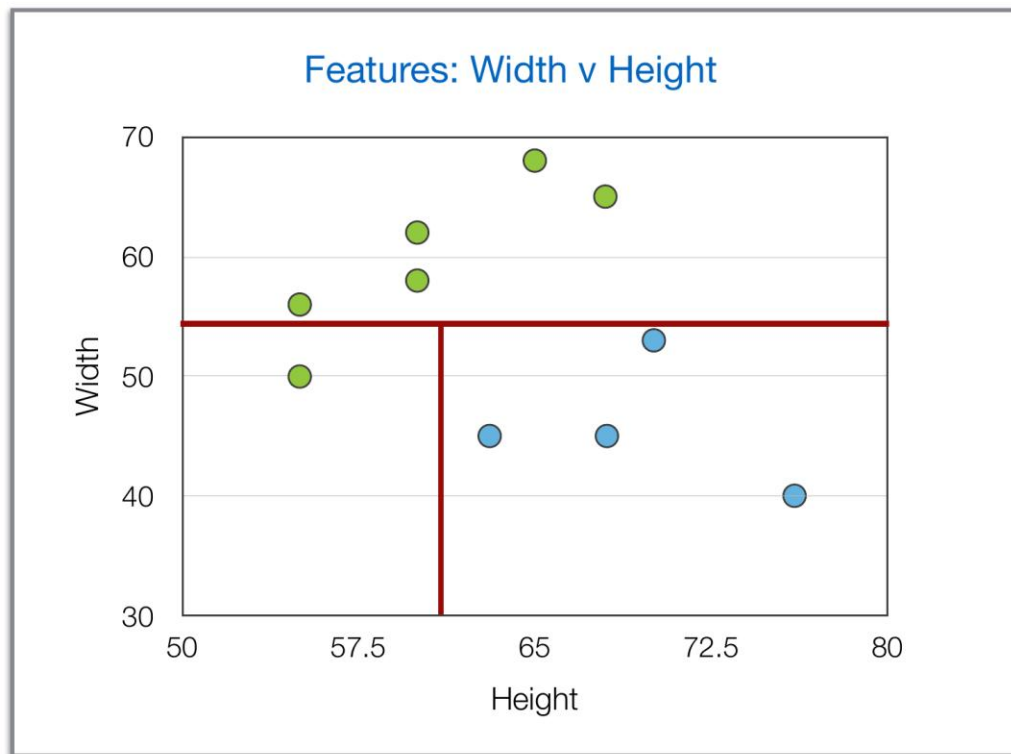
Example: Apples v Pears

- 10 training examples where each has a class label (“apple” or “pear”), and each is described with 6 features.

<i>Example</i>	<i>Colour</i>	<i>Height</i>	<i>Width</i>	<i>Taste</i>	<i>Weight</i>	<i>H/W</i>	<i>Class</i>
1	210	60	62	Sweet	186	0.97	Apple
2	220	70	53	Sweet	180	1.32	Pear
3	215	55	50	Tart	152	1.10	Apple
4	180	76	40	Sweet	152	1.90	Pear
5	220	68	45	Sweet	153	1.51	Pear
6	160	65	68	Sour	221	0.96	Apple
7	215	63	45	Sweet	140	1.40	Pear
8	180	55	56	Sweet	154	0.98	Apple
9	220	68	65	Tart	221	1.05	Apple
10	190	60	58	Sour	175	1.03	Apple

Example: Apples v Pears

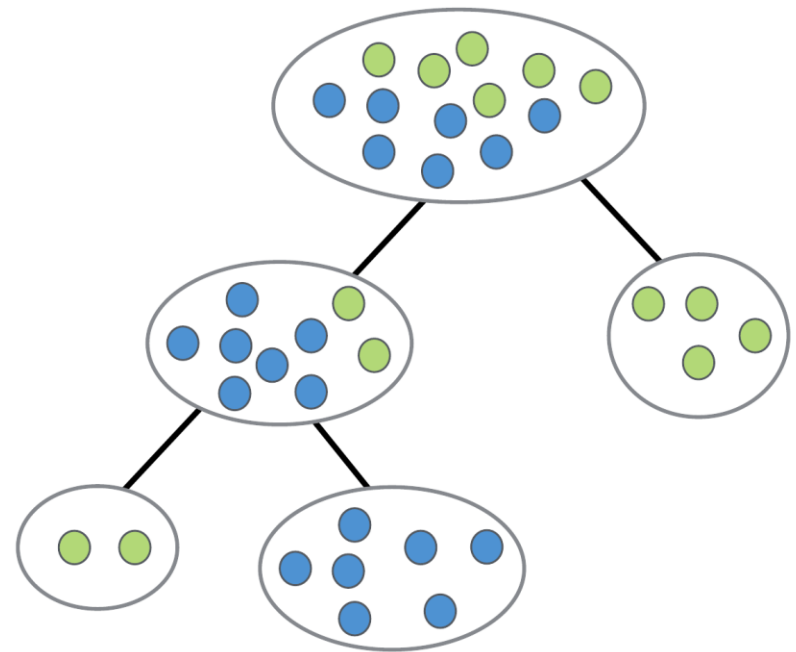
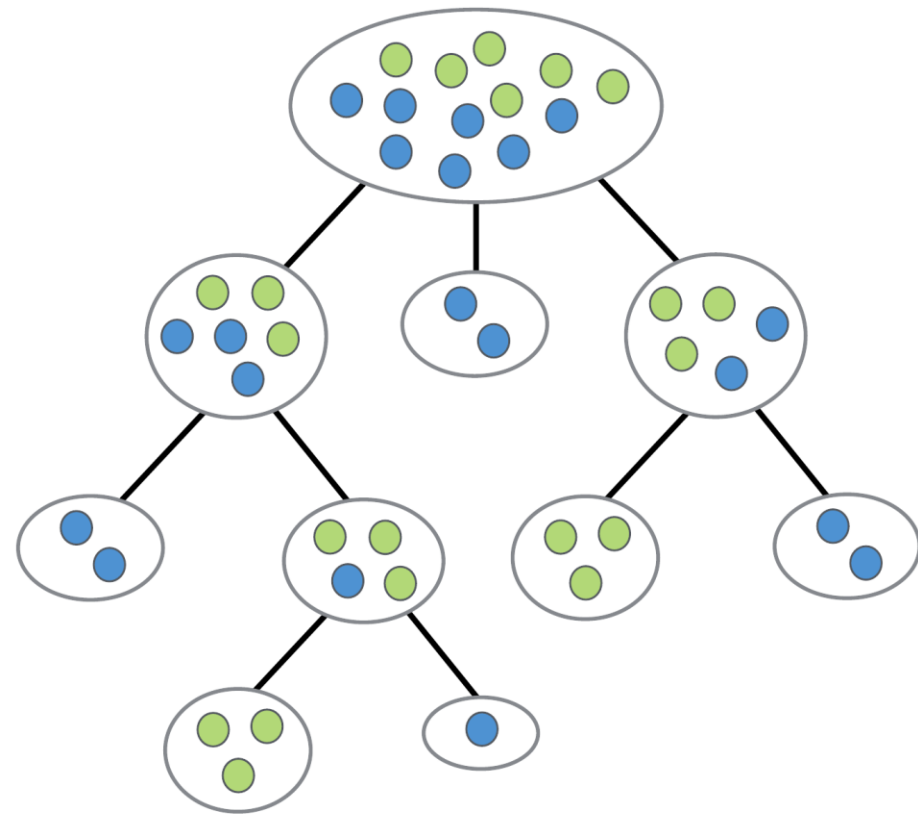
- Simple decision tree for classifying Apples v Pears using only 2 features: {Height, Weight}



Just 2 features can split the data based on these decision rules.

What makes a good tree?

Loads to choose from...



We prefer simpler trees...



Decision Trees Example

Q. “Will a customer wait for a restaurant table?”

Russell & Norvig, Artificial Intelligence: A Modern Approach, Prentice Hall, 2009.

Binary classification task (WillWait = {Yes, No}), with examples described by 10 descriptive features:

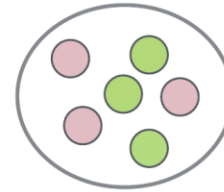
Example	Alternate	Bar	Fri/Sat	Hungry	Patrons	Price	Raining	Reservation	Type	WaitEst	WillWait?
1	Yes	No	No	Yes	Some	€€€	No	Yes	French	0-10	Yes
2	Yes	No	No	Yes	Full	€	No	No	Thai	30-60	No
3	No	Yes	No	No	Some	€	No	No	Burger	0-10	Yes
4	Yes	No	Yes	Yes	Full	€	No	No	Thai	10-30	Yes
5	Yes	No	Yes	No	Full	€€€	No	Yes	French	>60	No
6	No	Yes	No	Yes	Some	€€	Yes	Yes	Italian	0-10	Yes
7	No	Yes	No	No	None	€	Yes	No	Burger	0-10	No
8	No	No	No	Yes	Some	€€	Yes	Yes	Thai	0-10	Yes
9	No	Yes	Yes	No	Full	€	Yes	No	Burger	>60	No
10	Yes	Yes	Yes	Yes	Full	€€€	No	Yes	Italian	10-30	No
11	No	No	No	No	None	€	No	No	Thai	0-10	No
12	Yes	Yes	Yes	Yes	Full	€	No	No	Burger	30-60	Yes

➔ How do we build a “good” decision tree for this data set?

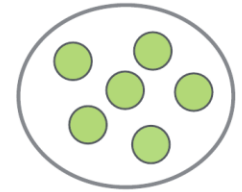


Entropy

- For decision trees, entropy provides a measure of impurity for each node in a tree - i.e. how uncertain we are about the decision for a given set of examples.



Node has high
uncertainty
→ High entropy



Node has low
uncertainty
→ Low entropy

- Calculating entropy:**

Entropy of a set of examples S with class labels $\{C_1, \dots, C_n\}$:

$$H(S) = \sum_{j=1}^n -p_j \log_2(p_j)$$

where p_j is the relative frequency (probability) of class C_j



Criterion: Information Gain

- **Information Gain (IG):** Popular information theoretic approach for selecting features in decision trees, based on entropy.
- Measures a feature's overall impact on entropy when used to split a set of training examples into two or more subsets.
 - How much information do we learn by splitting on the feature?
 - How much is the reduction in entropy?

- **Definition:**

IG for feature A that splits a set of examples S into $\{S_1, \dots, S_m\}$:

$$IG(S, A) = (\text{original entropy}) - (\text{entropy after split})$$

$$IG(S, A) = H(S) - \sum_{i=1}^m \frac{|S_i|}{|S|} H(S_i)$$

Each subset is weighted in proportion to its size



$$H(S) = - (0.5 \log 0.5 + 0.5 \log 0.5) = 1$$

$$H(\text{Hungry} = N) = - (1 \log 1) = 0 \quad (5/12 \text{个样本})$$

$$H(\text{Hungry} = Y) = - \left(\frac{2}{7} \log \frac{2}{7} + \frac{5}{7} \log \frac{5}{7} \right) = 0.259$$

(7/12个样本)

$$IG(\text{Hungry}) = H(S) - \frac{5}{12} H(\text{Hungry} = N) - \frac{7}{12}$$

$$H(\text{Hungry} = Y) = 0.849$$

计算其他特征的信息增益并选择增益最大的作为根节点

闭卷考试，好好复习
考试要带计算器
都能取得自己满意的成绩